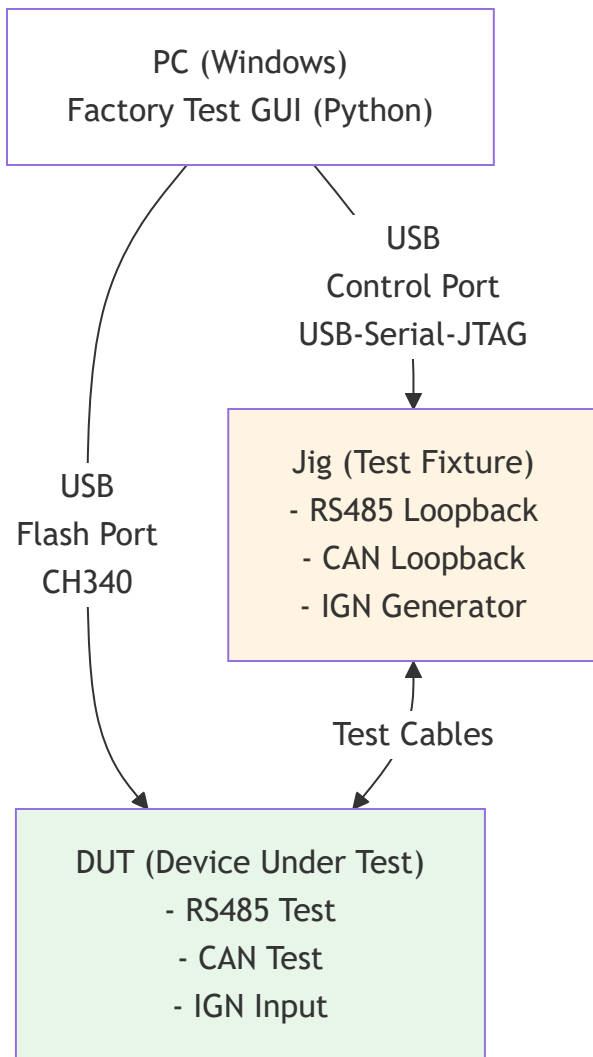


System Overview

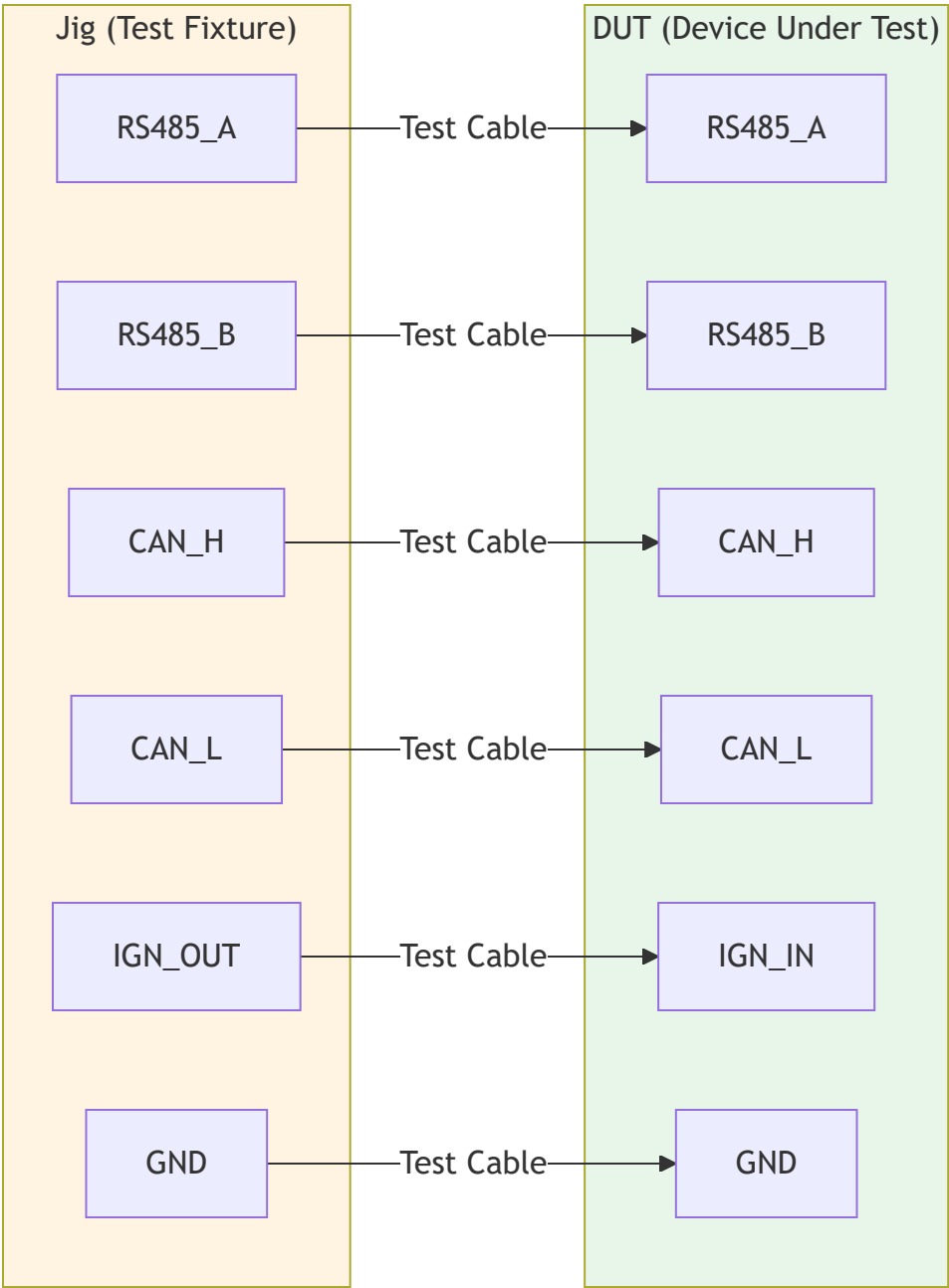


Components:

- **PC**: Runs test GUI software
- **Jig (Test Fixture)**: Provides RS485/CAN loopback and IGN signal
- **DUT (Device Under Test)**: PE board to be tested

Hardware Setup

1. Connect DUT (Jig ↔ DUT)



2. Connect USB Cables (PC ↔ Jig/DUT)

Device	USB Type	Port Identification	Purpose
Jig	USB-Serial-JTAG	"USB Serial" / "JTAG"	Control Port (Jig Control)
DUT	CH340	"CH340" / "CH343" / "CH9102"	Flash Port (Flashing/Testing)

3. Power Check

-  Jig powered normally (LED indicator)
-  DUT powered normally
-  All cables securely connected

Software Setup

Install Python Dependencies (First time only)

```
pip install PySide6 pyserial esptool
```

Detailed installation guide: [GUI_SETUP.md](#)

Build Firmware (Only when code updates)

```
cd pe_board  
idf.py build
```

Test Procedure

Test Flow Diagram

Control Port(Jig): <No Port>Flash Port(DUT): <No Port>Refresh PortsConnectDisconnect

Flash File: C:\Users\h1576\Desktop\Roam\pe_code\pe_board\build\flash_project_argsChange File

Progress: 0% - Ready

Test

Show Advanced

Selftest Results

OVERALL-

EG915-

IMEI: -

ICCID: -

Motion-

mag=0.000

RS485-

CAN-

GNSS UART-

EBL-V=0.00V

IGN Opto-

IM Opto-

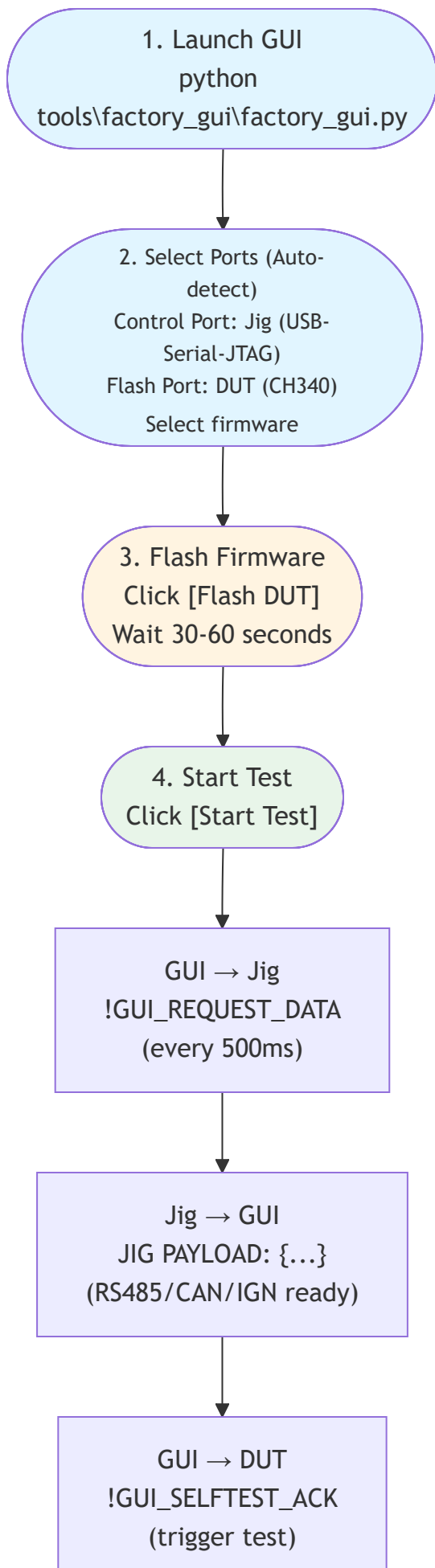
IBL-V=N/A

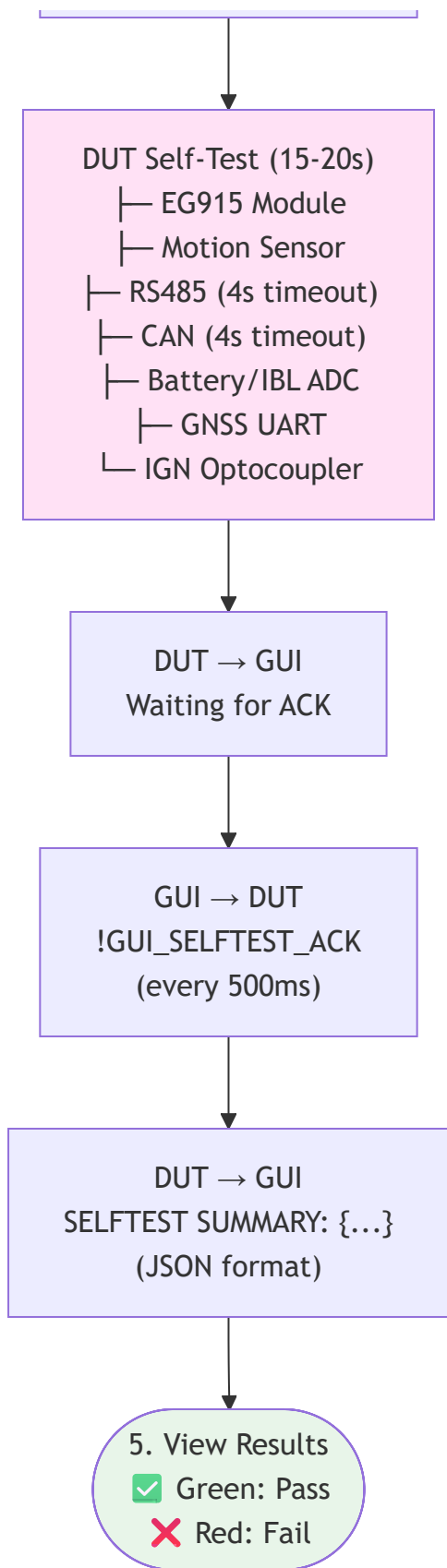
7-CH Voltage (K10-3U8)

	Channel	Voltage(V)	Status
1	AI1 (5V0_BUCK)	-	-
2	AI2 (5V0)	-	-
3	AI3 (3V3_AWS)	-	-
4	AI4 (4V0)	-	-
5	AI5 (3V3_LDO)	-	-
6	AI6 (4V7_BOS)	-	-
7	AI7 (3V3ANT)	-	-

Serial Log

[AUTO] No valid port found, please select and connect manually





Testing recodring

GUI → .csv file

名称

修改日期

类型

大小

.git

2025/11/21 10:44

文件夹

pycache

2025/11/21 10:44

文件夹

build

2025/11/21 10:49

文件夹

factory_gui.py

2025/11/18 17:32

Python.File

73 KB

factory_gui_config.json

2025/11/21 11:02

JSON 源文件

1 KB

GUI_SETUP.md

2025/11/19 19:18

Markdown 源文件

2 KB

README.md

2025/11/19 19:21

Markdown 源文件

4 KB

requirements.txt

2025/11/4 21:30

Text 源文件

1 KB

test_results.csv

2025/11/21 11:03

XLS 工作表

4 KB

	A	B	C	
1	Time	IMEI	ICCID	Over
2	#####	861076082:898822806(PAS		
3	#####	861076082:N/A	FAI	
4	#####	861076082:N/A	FAI	
5	#####	861076082:898822806(PAS		
6	#####	861076082:N/A	PAS	
7	#####	861076082:N/A	PAS	
8	#####	861076082:N/A	PAS	
9	#####	861076082:N/A	PAS	
10	#####	861076082:N/A	PAS	
11	#####	861076082:898822806(PAS		
12	#####	861076082:N/A	PAS	
13	#####	861076082:N/A	PAS	
14	#####	861076082:N/A	PAS	
15	#####	861076082:898822806(PAS		
16	#####	861076082:898822806(PAS		
17				
18				

test

Serial Command Protocol

GUI → Jig (Control Port)

Command: `!GUI_REQUEST_DATA`

Purpose: Request Jig to send test ready status

Frequency: Every 500ms (20s timeout)

Response: JIG PAYLOAD: `{"rs485_loopback": true, "can_loopback": true, ...}`

GUI → DUT (Flash Port)

Command: `!GUI_SELFTEST_ACK`

Purpose: ① Start DUT self-test ② Trigger result output

Frequency: Every 500ms (20s timeout)

Response: SELFTEST SUMMARY: `{"eg915_ok": true, "motion": {...}, ...}`

Test Items

Item	Test Content	Pass Criteria	Timeout
EG915	AT command comm., IMEI, ICCID	AT response OK	2s
Motion	Magnetometer reading	mag > 0.1	1s
RS485	Loopback test (send 8 bytes)	Receive 8 bytes match	4s
CAN	Loopback test (send 8 bytes)	Receive 8 bytes match	4s
Battery	Battery voltage ADC	0.441V ~ 0.539V	1s
IBL	IO2 voltage ADC	> 0 mV	1s
GNSS	UART data reception	> 0 bytes	800ms
IGN	Optocoupler level change	HIGH→LOW→HIGH	1s

Total Test Time: 15-25 seconds

Troubleshooting

✖ Port Not Detected

Symptom: No serial ports in dropdown menu

Solution:

1. Check USB cable connection
2. Click "Refresh Ports" button
3. Verify COM port in Device Manager
4. Install CH340 driver

✖ Flash Failed

Symptom: "Flash failed" error

Solution:

1. Confirm DUT is powered on
2. Replace USB cable (not charge-only cable)
3. Select correct Flash Port
4. Hold BOOT button and reconnect

✖ RS485/CAN Test Failed

Symptom: RS485 or CAN shows FAIL

Solution:

1. **If 4-second timeout:**
 - Check cable connections between Jig and DUT
 - Confirm GND is connected
 - Test if Jig is working properly
2. **If hardware damaged:**
 - DUT will still complete test after 4 seconds
 - Result shows FAIL but doesn't affect other tests

- Other test items execute normally

Test Timeout

Symptom: "Timeout waiting for test results"

Solution:

1. Confirm all test cables connected (RS485/CAN/IGN)
2. Confirm Jig is powered and running
3. Re-flash DUT firmware
4. Restart GUI and retry test